

Noise-Induced Localized Patterns in Excitable Media: Amplitude versus Persistence

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Transient spatially localized activity underlies a broad range of biological processes, yet the statistical property of fluctuations that controls its nucleation remains unclear. We systematically investigate noise-induced dynamics in a spatially extended FitzHugh-Nagumo excitable system and identify three regimes: a quiescent phase, a spatially extended alternating phase, and a pattern-forming phase characterized by transient localized excitation patches. Surprisingly, we find that temporal noise correlations are not required for patch formation: Gaussian white noise produces localized patches when its amplitude is sufficiently large, whereas weaker fluctuations can achieve the same effect when temporal correlations allow them to persist. These distinct mechanisms are linked through the integrated noise strength, which quantifies the accumulated stochastic forcing available to nucleate an excitation. The inhibitor response time subsequently determines whether the excitation remains localized or spreads throughout the system. Our results jointly identify the instantaneous amplitude and the persistence as the key stochastic control parameters for transient pattern formation in excitable media.

Introduction.—Biological functions emerge from complex networks of dynamically interacting molecules. In eukaryotic cell motility, the molecular networks that coordinate cell movement have been shown to exhibit the characteristic behavior of excitable reaction-diffusion systems [1–13]. These networks couple signaling proteins, including Ras and PI3K, to cytoskeletal components such as actin and its regulators. Intrinsic molecular fluctuations can stochastically activate the network, generating a rich variety of patterns, including propagating waves and transient, spatially localized patches of activity. An example is shown in Fig. 1, where PI3K activity at the basal surface of a cell was visualized using the fluorescence biosensor PH-AKT. Remarkably, even in the absence of external stimuli, the cell exhibited continual morphological changes accompanied by transient patches of PI3K activity that appeared spontaneously at different locations on the cell surface.

These spatiotemporal signaling patterns play essential roles in cellular function. Waves and localized patches of Ras-PI3K activity drive the formation of protrusions during cell movement [4, 5, 7, 11, 14, 15] and can trigger pulsatile activation of downstream pathways, including the extracellular signal-regulated kinase (ERK) pathway, a key regulator of cell proliferation [16–18]. These waves also coordinate the spatiotemporal dynamics of glycolytic enzymes involved in energy production [19]. Dysregulation of the underlying excitable network can disrupt both cell growth, motility, and metabolism. Moreover, activating mutations affecting the Ras-PI3K signaling network occur in more than 50% of human cancers [20] and have been associated with increased network excitability [14, 18, 19]. Hence, understanding the dynamical mechanisms that select among different signaling patterns is crucial for elucidating both normal cellular be-

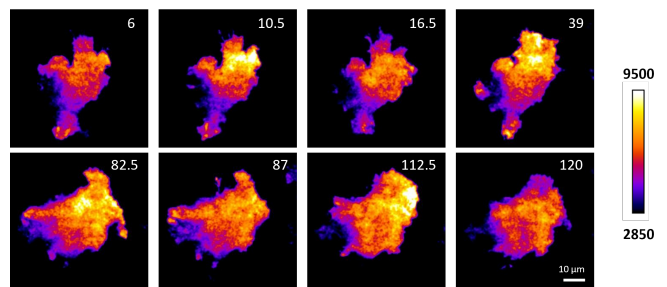


FIG. 1. Time-lapse images of a 4T1 tumor cell expressing the biosensor PH-AKT, which detects the activity of phosphoinositide 3-kinase (PI3K) [21]. The images were acquired using total internal reflection fluorescence microscopy, which captures fluorescence signals very close (< 200 nm) to the cover slip under the basal cell surface. Time stamp: min. Color scale: fluorescence intensity (arbitrary units).

havior and its pathological disruption.

These observations raise a broader question: what properties of stochastic fluctuations determine whether an excitable medium remains quiescent, develops system-wide activity, or sustains production of transient localized patterns? In particular, is nucleation controlled by the instantaneous noise amplitude, the duration over which a fluctuation persists, or by the total stochastic forcing accumulated over time? To address these questions, we consider the FitzHugh-Nagumo (FHN) model. Originally introduced as a reduced description of excitation and recovery in nerve membranes [22, 23] and now widely used as a paradigmatic model of excitable dynamics, the model couples a rapidly responding activator to a slower inhibitory recovery variable. Once the activator crosses an excitation threshold, nonlinear amplification drives it toward an excited state, after which the

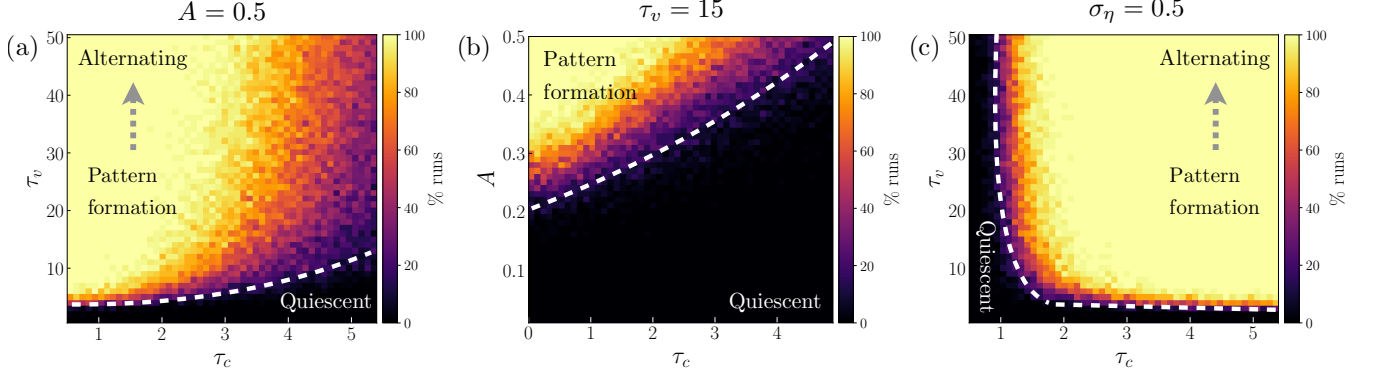


FIG. 2. Noise-induced dynamical phases under three parameterizations of the active noise. Color indicates the percentage of 50 independent runs exhibiting noise-induced excitation. Panel (a) is obtained by varying the inhibitor response time $1 \leq \tau_v \leq 50$ and correlation time $0.5 \leq \tau_c < 5.5$ at fixed integrated noise strength $A = 0.5$. Panel (b) is obtained by sweeping the ranges $0.01 \leq A \leq 0.5$ and $0 \leq \tau_c \leq 4.9$ at fixed $\tau_v = 15$. Panel (c) is obtained by varying $1 \leq \tau_v \leq 50$ and $0.5 \leq \tau_c < 5.5$ at fixed equal-time rms noise amplitude $\sigma_\eta = 0.5$. The white dashed curves delineate the boundary of the quiescent regime, while the arrows in (a) and (c) indicate the crossover from localized pattern formation to the spatially extended alternating phase. The remaining parameters are $\alpha = 0.5$, $\beta = 0.7$, $D_u = 1$, and $D_v = 5$.

inhibitor terminates the excitation and restores the resting state. Promoting these variables to spatially varying fields and introducing diffusive coupling yields a minimal reaction-diffusion description of an excitable medium capable of supporting propagating waves and other spatially organized structures [23–25]. Intrinsic and environmental fluctuations can then be incorporated through a stochastic Langevin force, conventionally represented by Gaussian white noise [26].

Although white noise is mathematically convenient, fluctuations in physical, chemical, and biological systems often develop over finite times and hence possess nontrivial temporal correlations [27–30]. Such persistent fluctuations arise naturally in nonequilibrium and active systems and are commonly modeled using colored noise [31–37]. Temporal correlations have been shown to strongly influence noise-induced dynamics in FitzHugh–Nagumo and other excitable systems [25, 38–40], as well as wave nucleation and pattern formation in spatially extended reaction–diffusion systems [41–46]. Finite-duration forcing has also been invoked to generate transient localized structures, partly because strong white noise was expected to obscure coherent spatial patterns [47, 48].

Here, we systematically interpolate between white and temporally correlated Gaussian noise to identify the key drivers for pattern formation. We find that transient localized patches can be nucleated either by short-lived fluctuations with sufficiently large instantaneous amplitude or by weaker fluctuations that persist for longer durations. These two routes are unified by the integrated noise strength, which measures the stochastic forcing accumulated over time. The inhibitor response time then determines whether a nucleated excitation is suppressed locally or spreads into system-wide alternating activity. The instantaneous amplitude as well as the temporal persistence emerge on equal footing as central stochastic

quantities governing the formation of transient localized patterns.

Reaction-diffusion equations.—To isolate the essential ingredients underlying stochastic pattern formation, we consider a minimal one-dimensional FitzHugh–Nagumo reaction-diffusion system defined on a circle. We denote the spatial coordinate by ℓ and impose periodic boundary conditions. The activator and inhibitor fields, $u(\ell, t)$ and $v(\ell, t)$, evolve according to

$$\frac{\partial u}{\partial t} = D_u \partial_\ell^2 u + \frac{1}{\tau_u} (1 - u^2)(u - v) + \eta(\ell, t), \quad (1a)$$

$$\frac{\partial v}{\partial t} = D_v \partial_\ell^2 v + \frac{1}{\tau_v} (u - \alpha v + \beta). \quad (1b)$$

Here, D_u and D_v are the diffusion coefficients of the activator and inhibitor, while τ_u and τ_v set their characteristic response times. The parameters α and β determine the relaxation and homogeneous steady-state value of the inhibitor. We focus on the regime $D_v > D_u$, in which the inhibitor diffuses more rapidly than the activator, and take $\alpha < \beta$. In the absence of noise, the system has a quiescent homogeneous state given by $u_0 = -1$ and $v_0 = (\beta - 1)/\alpha$. Stochastic fluctuations enter Eq. (1a) through the additive noise field $\eta(\ell, t)$ acting on the activator.

Ornstein-Uhlenbeck active noise.—To investigate how temporal persistence affects the dynamics of the excitable medium, we model $\eta(\ell, t)$ as an Ornstein-Uhlenbeck process with zero mean and autocovariance

$$\langle \eta(\ell, t) \eta(\ell', t') \rangle = \delta(\ell - \ell') \sigma_\eta^2 e^{-|t - t'|/\tau_c}, \quad (2)$$

where σ_η denotes the equal-time root mean square (rms) amplitude of the stochastic forcing and τ_c is the temporal correlation time. An important quantity, which will emerge as a key quantity to unify the pattern-formation

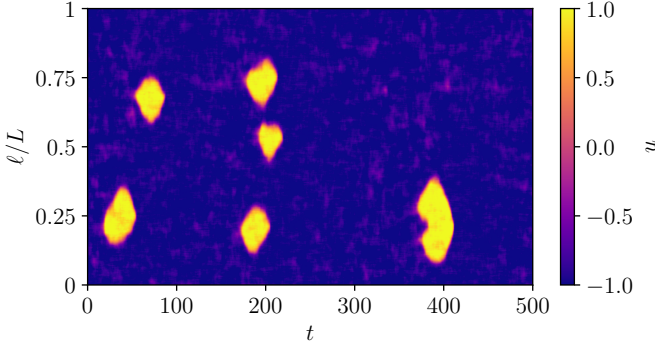


FIG. 3. Spacetime diagram of the activator field $u(\ell, t)$, illustrating the spontaneous emergence of transient, spatially localized excitation patches. Time is plotted horizontally, while the vertical axis gives the normalized length ℓ/L along the periodic domain; the color scale indicates the local activator value. Starting from the quiescent background $u \simeq -1$, localized excited regions ($u \simeq 1$) nucleate at different positions and times, evolve over finite lifetimes, and eventually decay. The parameters are $\alpha = 0.5$, $\beta = 0.7$, $\tau_v/\tau_u = 25$, $\tau_c/\tau_u = 5$, $\Delta t/\tau_u = 0.1$, and $A = 1.25$.

mechanisms is the *integrated noise strength*

$$A \equiv \int_0^\infty dt \langle \eta(\ell, 0) \eta(\ell, t) \rangle = \sigma_\eta^2 \tau_c, \quad (3)$$

which measures the total stochastic forcing accumulated over the correlation time. Henceforth, we regard A , rather than σ_η , as the primary control parameter.

Equivalently, the noise field evolves according to

$$\frac{\partial \eta(\ell, t)}{\partial t} = -\frac{\eta(\ell, t)}{\tau_c} + \frac{\sqrt{2A}}{\tau_c} \xi(\ell, t), \quad (4)$$

where $\xi(\ell, t)$ is a Gaussian white noise field satisfying $\langle \xi(\ell, t) \rangle = 0$ and $\langle \xi(\ell, t) \xi(\ell', t') \rangle = \delta(\ell - \ell') \delta(t - t')$. To simulate the nonlinear FitzHugh-Nagumo reaction-diffusion equations with active noise, we employ the Crank-Nicolson method; details of the numerical implementation are presented in Appendix A.

Dynamical phases.—Figure 2 summarizes the noise-induced dynamics of the stochastic FHN system. Figs. 2(a) and (c) show phase diagrams obtained by varying the inhibitor response time τ_v and the noise correlation time τ_c at fixed integrated noise strength A and fixed equal-time rms amplitude σ_η , respectively. Figure 2(b) shows the corresponding dependence on A and τ_c at fixed τ_v . Each point represents the cumulative data from 50 runs, each of duration $500\tau_u$. Together, these complementary parameterizations disentangle the roles of inhibitor dynamics, noise amplitude, and temporal persistence in controlling the emergence of localized excitation patches. Three dynamical regimes emerge: a quiescent phase, a spatially extended alternating phase, and an intermediate pattern-forming phase characterized by transient, spatially localized excitation patches. We now discuss the physical mechanisms underlying each regime.

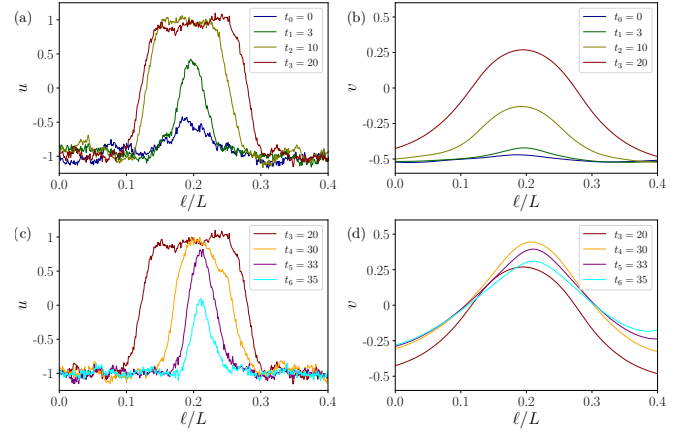


FIG. 4. Spatial profiles of the activator u and inhibitor v over the lifetime of a single excitation patch. Panels (a) and (b) show its noise-induced formation and rapid growth, while panels (c) and (d) show its contraction and decay due to the delayed inhibitor response. Each curve corresponds to a specified time offset.

In the quiescent phase, the activator remains near its homogeneous steady-state value $u_0 = -1$, apart from small fluctuations. For $-1 < u < 1$, the local reaction term in Eq. (1a) amplifies the activator only when $u > v$, which near equilibrium corresponds approximately to crossing the threshold $v_0 = (\beta - 1)/\alpha$. The system remains quiescent when fluctuations are either too weak or too short-lived to overcome this threshold and initiate nonlinear amplification. Quiescence also occurs when the inhibitor responds on a timescale comparable to or shorter than that of the activator, $\tau_v/\tau_u \lesssim 1$, because the inhibitory feedback suppresses an incipient excitation before it can develop.

At the opposite extreme, a sufficiently slow inhibitor produces a spatially extended alternating phase. When $\tau_v/\tau_u \gg 1$, the inhibitor cannot arrest the growth of an excitation before it spreads throughout the system. The activator consequently approaches an approximately uniform excited state with $u \simeq 1$. As the inhibitor gradually accumulates, it suppresses the activity and restores the quiescent state, after which the cycle repeats once the inhibitor relaxes. The oscillation period is therefore set primarily by the inhibitor response time τ_v .

The pattern-forming phase lies between these two limits and is the principal focus of this work. Figure 3 shows the spatiotemporal evolution of the activator field $u(\ell, t)$ in the pattern-forming regime. Localized regions with $u \simeq 1$ nucleate spontaneously at random positions against the quiescent background, persist for a finite time, and subsequently disappear, after which new patches may emerge elsewhere. Individual nucleation events exhibit similar spatial extent and lifetime, although larger patches occasionally arise through the merger of neighboring excitations.

The activator and inhibitor profiles in Fig. 4 illustrate the evolution of an individual patch. A sufficiently

strong fluctuation first drives the activator above the local excitation threshold, after which the nonlinear reaction term rapidly amplifies the excitation. Because the inhibitor responds more slowly, the patch is able to grow for a finite interval before inhibitory feedback accumulates, raises the excitation threshold, and terminates the activity. Localized pattern formation therefore requires a separation of timescales: the inhibitor must respond slowly enough to permit nucleation and amplification, yet rapidly enough to arrest the excitation before it becomes system-wide.

Figure 2(a) shows how the inhibitor dynamics interact with the stochastic forcing at fixed integrated noise strength A . Since $\sigma_\eta^2 = A/\tau_c$, increasing τ_c has two competing effects: it increases the temporal persistence of individual fluctuations while simultaneously reducing their equal-time rms amplitude. Temporal persistence favors nucleation by allowing a fluctuation to act coherently over a longer interval, whereas the reduced instantaneous amplitude makes it less likely for the activator to cross the excitation threshold. At fixed A , the latter effect eventually dominates as τ_c increases. Consequently, sustaining localized pattern formation at larger τ_c requires a slower inhibitor response, producing the upward-sloping quiescent–pattern boundary in Fig. 2(a).

Figure 2(b) further demonstrates how the integrated noise strength A and correlation time τ_c jointly control excitation nucleation. For fixed τ_v and τ_c , increasing A drives the system from the quiescent phase into the localized pattern-forming phase. The critical value of A increases monotonically with τ_c , consistent with the same competition seen in Fig. 2(a): at fixed A , longer-lived fluctuations have a smaller equal-time rms amplitude and therefore require a larger integrated noise strength to achieve nucleation. Once an excitation is nucleated, however, its subsequent evolution depends only weakly on A , indicating that its growth and decay are governed primarily by the intrinsic reaction-diffusion dynamics and, in particular, by the inhibitor response time.

Importantly, Fig. 2(b) also shows that temporal correlations are not required for localized pattern formation. The pattern-forming regime extends continuously to the white noise limit $\tau_c = 0$, demonstrating that Gaussian white noise alone can nucleate transient excitation patches when its strength is sufficiently large. In this regime, Gaussian white noise shifts the mean values of u and v away from their deterministic quiescent values, producing a nonequilibrium quiescent background. Fluctuations about this background must drive u above the corresponding nonequilibrium inhibitor level before nonlinear amplification takes over. Once this occurs, localized excitations remain robust in the presence of the white noise fluctuations. The nonequilibrium quiescent background can be characterized analytically within a mean-field treatment; see Appendix B for details.

The complementary parameterization in Fig. 2(c) provides striking direct evidence for the role of temporal persistence. Here, the equal-time rms amplitude is held

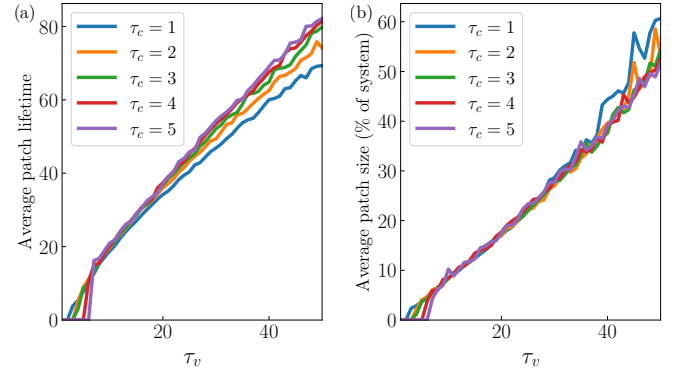


FIG. 5. (a) Patch lifetime (in units of τ_u) and (b) patch size (percentage of system size) as functions of τ_v for fixed values of $\tau_c = 1, 2, 3, 4, 5$. The plots indicate that both patch lifetime and size scale linearly with τ_v .

fixed at $\sigma_\eta = 0.5$, so increasing τ_c increases the duration of the fluctuations without reducing their instantaneous amplitude. The resulting behavior is qualitatively opposite to that in Fig. 2(a). Increasing τ_c now strongly promotes localized pattern formation, and the quiescent region rapidly contracts as the fluctuations become more persistent. Thus, fluctuations that are insufficient to nucleate an excitation when short-lived can cross the effective excitation threshold when they persist for a sufficiently long time.

The contrasting trends in Figs. 2(a) and (c) distinguish the effects of noise amplitude and persistence. At fixed A , increasing τ_c necessarily reduces σ_η and ultimately suppresses nucleation; at fixed σ_η , increasing τ_c enhances persistence without this amplitude penalty and instead promotes nucleation. Together, these results show that temporal correlation is neither intrinsically detrimental nor essential to pattern formation. Rather, its effect depends on how the noise is parameterized: both the instantaneous amplitude of a fluctuation and the time over which it persists determine whether sufficient stochastic forcing accumulates to trigger nonlinear excitation.

We emphasize, however, that the fixed- σ_η parameterization of Fig. 2(c) does not provide a continuous interpolation to Gaussian white noise. Since $A = \sigma_\eta^2 \tau_c$, the integrated noise strength vanishes as $\tau_c \rightarrow 0$ at fixed σ_η . The white-noise limit with finite stochastic forcing must instead be taken at fixed A , for which $\sigma_\eta^2 = A/\tau_c$ diverges as $\tau_c \rightarrow 0$. Therefore, quantitative comparisons between colored and Gaussian white noise should be made using Figs. 2(a) and (b), whereas Fig. 2(c) isolates the physical effect of increasing temporal persistence at fixed instantaneous noise amplitude.

The characteristic spatial extent of a patch can be estimated from straightforward dimensional analysis. Once the patch is formed, the inhibitor rises to extinguish it on the timescale of τ_v . During this interval, the activator front propagates spatially at a speed of order $\sqrt{D_u/\tau_u}$ while the inhibitor diffuses at a speed $\sqrt{D_v/\tau_v}$. The char-

characteristic length scales for the activator and inhibitor are the products of the propagation speeds and τ_v , given by

$$\ell_u \sim \tau_v \sqrt{D_u/\tau_u} \quad \text{and} \quad \ell_v \sim \sqrt{D_v \tau_v}, \quad (5)$$

respectively. Here, ℓ_u estimates the distance traveled by the activator front before inhibitory suppression, whereas ℓ_v characterizes the spatial spread of the inhibitor. These estimates are consistent with the profiles in Fig. 4. Since the nonlinear growth of the activator occurs rapidly, τ_v dominates as the characteristic timescale. See Appendix C for more details about the properties of the patches over the ranges swept by the phase diagram in Fig. 2.

Discussion.—In this work, we investigated noise-induced pattern formation in a stochastic FitzHugh-Nagumo reaction-diffusion system driven by temporally correlated fluctuations. We identified three dynamical regimes, the most intriguing of which being a pattern-forming phase consisting of transient localized excitation patches. Our central results illustrate that localized patches are nucleated either by strong, short-lived fluctuations or by weaker fluctuations that persist for longer times. The persistence of the fluctuations and the instantaneous fluctuation amplitude emerge on even footing as critical stochastic control parameters governing excitation nucleation. Once nucleated, the subsequent growth, lifetime, and eventual suppression of an excitation are governed predominantly by the inhibitor response time.

These findings provide a simple physical framework for understanding stochastic pattern formation in excitable biochemical signaling networks. In particular, the results suggest that the morphology of transient Ras-PI3K signaling patches depend jointly on the amplitude as well as persistence of intracellular biochemical fluctuations, along with the characteristic recovery time of the inhibitory network. This perspective may help explain how cells exhibiting different sources of molecular fluctuations can nevertheless display similar protrusive dynamics. More broadly, the phase diagrams presented here provide experimentally testable predictions for how changes in signaling noise or inhibitor kinetics regulate the transition between quiescent, localized, and system-wide activity. A potential application of these findings in cell manipulation or therapy is that relatively weak but persistent perturbations may be used to drive specific excitable systems across the excitation threshold, potentially reducing unwanted effects on other systems associated with stronger stimulation.

Beyond cell signaling, we expect our results to apply more broadly to stochastic excitable media in which fluctuations possess finite temporal correlations. Examples include intracellular biochemical networks, active matter systems, ecological and epidemiological reaction-diffusion models, and other nonlinear systems exhibiting threshold dynamics. It would also be interesting to examine whether this principle extends to higher-dimensional excitable media, more realistic signaling networks, and active systems with non-Gaussian or spatially correlated fluctuations.

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Appendix A: Crank-Nicolson method

In this Appendix, we describe, for pedagogical purposes, the numerical implementation used to solve the nonlinear stochastic FitzHugh-Nagumo reaction-diffusion equations, Eq. (1), driven by the temporally correlated noise $\eta(\ell, t)$ characterized by Eq. (2). An equivalent description of the correlated noise is provided by the stochastic differential equation Eq. (4), in which η evolves under a Langevin white noise ξ . For the numerical simulations, Eq. (4) is discretized using the Euler-Maruyama scheme,

$$\eta_i(t + \Delta t) = \left(1 - \frac{\Delta t}{\tau_c}\right) \eta_i(t) + \frac{\sqrt{2A\Delta t}}{\tau_c} \xi_i(t), \quad (\text{A1})$$

where i labels the spatial grid points and each $\xi_i(t)$ is an independent standard Gaussian random variable. The factor $\sqrt{\Delta t}$ ensures the correct continuum-time scaling. At each time step, the updated noise field $\eta_i(t)$ enters the activator dynamics through Eq. (1a).

The FHN equations are advanced in time using the Crank-Nicolson method, an implicit scheme that is second-order accurate in time. We adopt this method because it is unconditionally stable for conventional *linear* diffusion equations, whereas explicit methods require substantially smaller time steps to maintain stability at the same spatial resolution. The principal complication arises from the nonlinear reaction term in the activator equation, Eq. (1a), which we write as

$$\varphi(\ell, t) \equiv \frac{1}{\tau_u} (1 - u^2)(u - v). \quad (\text{A2})$$

The nonlinear dependence of φ on u and v requires the solution of a nonlinear system at every time step. Although one could simplify the problem by linearizing φ , such an approximation is only appropriate when the fluctuations of u and v remain small. Instead, we retain the full nonlinearity and determine the solution using a fixed-point iteration. Consequently, the time step must be chosen sufficiently small to resolve the reaction dynamics, ensure convergence of the nonlinear iteration, and accurately integrate the stochastic noise process.

The periodic spatial domain is discretized into N uniformly spaced grid points,

$$\ell_i = ia, \quad a = L/N, \quad (\text{A3})$$

for $i = 0, \dots, N-1$, with all spatial indices understood modulo N . Let $\mathbf{L}$ denote the standard second-order finite-difference Laplacian with periodic boundary conditions. At time $t_n = n\Delta t$, the discretized fields are denoted by $\mathbf{u}^n$ and $\mathbf{v}^n$, and the nonlinear reaction term is evaluated componentwise as

$$\varphi_i^n = \frac{1}{\tau_u} [1 - (u_i^n)^2] (u_i^n - v_i^n). \quad (\text{A4})$$

Applying the Crank-Nicolson discretization to the activator and inhibitor equations yields

$$\frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta t} = \frac{D_u}{2} \mathbf{L} (\mathbf{u}^{n+1} + \mathbf{u}^n) + \frac{\varphi^{n+1} + \varphi^n}{2}, \quad (\text{A5})$$

$$\begin{aligned} \frac{\mathbf{v}^{n+1} - \mathbf{v}^n}{\Delta t} &= \frac{D_v}{2} \mathbf{L} (\mathbf{v}^{n+1} + \mathbf{v}^n) \\ &+ \frac{1}{2\tau_v} [\mathbf{u}^{n+1} + \mathbf{u}^n - \alpha (\mathbf{v}^{n+1} + \mathbf{v}^n) + 2\beta \mathbf{1}], \end{aligned} \quad (\text{A6})$$

where $\mathbf{1}$ denotes the constant unit vector. Introducing the coefficients

$$\lambda_u = \frac{D_u \Delta t}{2}, \quad \lambda_v = \frac{D_v \Delta t}{2}, \quad c_v = \frac{\Delta t}{2\tau_v}, \quad (\text{A7})$$

the corresponding system matrices become

$$\mathbf{A}_u = \mathbf{I} - \lambda_u \mathbf{L}, \quad \mathbf{A}_v = \mathbf{I} - \lambda_v \mathbf{L} + \alpha c_v \mathbf{I}, \quad (\text{A8})$$

$$\mathbf{B}_u = \mathbf{I} + \lambda_u \mathbf{L}, \quad \mathbf{B}_v = \mathbf{I} + \lambda_v \mathbf{L} - \alpha c_v \mathbf{I}. \quad (\text{A9})$$

The coupled equations can then be written compactly as

$$\begin{aligned} \begin{bmatrix} \mathbf{A}_u & \mathbf{0} \\ -c_v \mathbf{I} & \mathbf{A}_v \end{bmatrix} \begin{bmatrix} \mathbf{u}^{n+1} \\ \mathbf{v}^{n+1} \end{bmatrix} \\ = \begin{bmatrix} \mathbf{B}_u & \mathbf{0} \\ c_v \mathbf{I} & \mathbf{B}_v \end{bmatrix} \begin{bmatrix} \mathbf{u}^n \\ \mathbf{v}^n \end{bmatrix} + \begin{bmatrix} \frac{\Delta t}{2} (\varphi^n + \varphi^{n+1}) \\ 2\beta c_v \mathbf{1} \end{bmatrix}. \end{aligned} \quad (\text{A10})$$

The coefficient matrices on the left-hand side, as well as the matrix multiplying the previous time step solution on the right-hand side, are independent of time and therefore need to be constructed only once at the beginning of the simulation.

The remaining challenge is that φ^{n+1} depends nonlinearly on the unknown fields $\mathbf{u}^{n+1}$ and $\mathbf{v}^{n+1}$. We therefore solve Eq. (A10) using a fixed-point iteration. The iteration is initialized with

$$\varphi^{n+1,0} = \varphi^n, \quad (\text{A11})$$

and, given the nonlinear term from the k th iteration, the updated fields are obtained from

$$\begin{aligned} \begin{bmatrix} \mathbf{A}_u & \mathbf{0} \\ -c_v \mathbf{I} & \mathbf{A}_v \end{bmatrix} \begin{bmatrix} \mathbf{u}^{n+1,k+1} \\ \mathbf{v}^{n+1,k+1} \end{bmatrix} \\ = \begin{bmatrix} \mathbf{B}_u & \mathbf{0} \\ c_v \mathbf{I} & \mathbf{B}_v \end{bmatrix} \begin{bmatrix} \mathbf{u}^n \\ \mathbf{v}^n \end{bmatrix} + \begin{bmatrix} \frac{\Delta t}{2} (\varphi^n + \varphi^{n+1,k}) \\ 2\beta c_v \mathbf{1} \end{bmatrix}. \end{aligned} \quad (\text{A12})$$

The nonlinear reaction term is then updated according to

$$\varphi_i^{n+1,k+1} = \frac{1}{\tau_u} \left[1 - (u_i^{n+1,k+1})^2 \right] (u_i^{n+1,k+1} - v_i^{n+1,k+1}), \quad (\text{A13})$$

and the iteration is repeated until the residual

$$\|\varphi^{n+1,k+1} - \varphi^{n+1,k}\| \quad (\text{A14})$$

falls below a prescribed tolerance, or until a specified maximum number of iterations is reached. Once the solution at the new time step has converged, the stochastic forcing is incorporated by updating the activator field using the Euler-Maruyama noise increment given in Eq. (A1).

Appendix B: Nonequilibrium quiescent background

In the presence of Gaussian white noise, we find that the mean values of the activator and inhibitor values are shifted compared to their quiescent values in the absence of noise, $u_0 = -1$ and $v_0 = (\beta - 1)/\alpha$. A nonequilibrium quiescent background emerges, with activator and inhibitor values now fluctuating near new nonequilibrium quiescent means μ_u and μ_v , respectively. Fluctuations must excite u above the new nonequilibrium inhibitor mean μ_v for the nonlinear amplification process to take over and nucleate a localized pattern. In this Appendix, we discuss how to analytically determine the nonequilibrium quiescent means, and predict if the integrated noise strength A is sufficiently strong for the system to enter the pattern-forming regime.

While the full FitzHugh-Nagumo equations cannot be fully analytically solved, applying some simple approximations allows us to analytically characterize the behavior of the nonequilibrium quiescent state. These approximations are motivated by observations of the simulations. Noting that the noise is spatially homogeneous, we perform spatial averaging over the periodic 1D system, where henceforth we use $\bar{X} \equiv L^{-1} \oint d\ell X(\ell)$. Applying the averaging to both sides of the reaction-diffusion equation subject to Gaussian white noise, we find

$$\frac{\partial \bar{u}}{\partial t} = \frac{1}{\tau_u} (\bar{u} - \bar{v} - \bar{u}^3 + \bar{u}^2 \bar{v}) \quad (\text{B1})$$

$$\frac{\partial \bar{v}}{\partial t} = \frac{1}{\tau_v} (\bar{u} - \alpha \bar{v} + \beta) \quad (\text{B2})$$

Here, the diffusive terms are eliminated because the averaging is done over a spatially periodic system and $\bar{\eta} \approx 0$ because the noise has zero mean.

In the nonequilibrium quiescent background, the spatial averages $\bar{u}$ and $\bar{v}$ are time-independent, and are denoted by μ_u and μ_v , respectively. Moreover, in the absence of noise they revert to $\mu_u = u_0$ and $\mu_v = v_0$. Approximating that $u(\ell)$ is spatially Gaussian distributed

and neglecting the covariance between u^2 and v , the equations of motion reduce to

$$0 \approx \mu_u - \mu_v - \mu_u^3 - 3\mu_u\sigma_u^2 + (\mu_u^2 + \sigma_u^2)\mu_v, \quad (\text{B3a})$$

$$0 \approx \mu_u - \alpha\mu_v + \beta \quad (\text{B3b})$$

for the nonequilibrium quiescent background. The coupled equations above relate the nonequilibrium quiescent background activator and inhibitor averages, μ_u and μ_v , to σ_u^2 .

The variance σ_u^2 can be estimated in a straightforward linearized treatment. To this end, we first begin by expanding the values of $u(\ell)$ and $v(\ell)$ around μ_u and μ_v ,

$$u(\ell) = \mu_u + \delta u(\ell), \quad v(\ell) = \mu_v + \delta v(\ell). \quad (\text{B4})$$

The latter will be solved for self-consistently, upon including Eq. (B3). Since μ_u is equated to the steady-state spatial average of u , the spatial average of the fluctuations δu is zero. Moreover, through empirical observation of simulations of the nonequilibrium quiescent background, we find that the fluctuations $\delta v/\mu_v \ll 1$ are small. Hence, when we linearize the equation of motion for activator u , we can neglect contributions proportional to δv , yielding

$$\partial_t \delta u = D_u \partial_\ell^2 \delta u + \tau_u^{-1} (1 - 3\mu_u^2 + 2\mu_u\mu_v) \delta u + \eta. \quad (\text{B5})$$

Applying a discrete spatial Fourier transform, we can make the substitution $\partial_\ell^2 \delta u \rightarrow -2a^{-2} (1 - \cos \frac{2\pi m}{N}) \delta u_m$, where δu_m is the amplitude of each Fourier mode. The governing linearized equation for the modes becomes

$$\partial_t \delta u_m = -r_m \delta u_m + \eta_m, \quad (\text{B6a})$$

where

$$r_m = \frac{3\mu_u^2 - 2\mu_u\mu_v - 1}{\tau_u} + \frac{2D_u}{a^2} \left(1 - \cos \frac{2\pi m}{N} \right) \quad (\text{B6b})$$

is the damping coefficient for Fourier mode m . Examining the damping coefficient, we see that the first term suppresses long wavelength fluctuations over the time scale of τ_u while short wavelength fluctuations are suppressed over the diffusion time scale a^2/D_u .

Time discretization is performed by using the Crank-Nicolson method. Doing so, we arrive at the implicit equation

$$\delta u_m^{n+1} = \frac{1 - r_m \Delta t/2}{1 + r_m \Delta t/2} \delta u_m^n + \sqrt{2A\Delta t} \xi_m^n, \quad (\text{B7})$$

where ξ_m^n is the Gaussian white noise sample driving mode m at time step n and A is the integrated noise strength defined in Eq. (3). We expect that the fluctuations δu_m^n are independent of the time step n . These statistical properties can be quantified by the *ensemble variance* of the mode m via $\langle |\delta u_m^n|^2 \rangle$, where the averaging $\langle \dots \rangle$ is done over an ensemble of simulations at time step n . Hence, we expect that for self-consistency, $\langle |\delta u_m^n|^2 \rangle = \langle |\delta u_m^{n'}|^2 \rangle$ for all values of n and n' .

To find the ensemble variance for the mode m , we square both sides of Eq. (B7) and perform the ensemble average. We apply the stationary condition $\langle |\delta u_m^n|^2 \rangle = \langle |\delta u_m^{n+1}|^2 \rangle$. Finally, we arrive at the expression for the stationary ensemble variance of mode m ,

$$\langle |\delta u_m|^2 \rangle = \frac{A}{r_m} \left(1 + \frac{r_m \Delta t}{2} \right)^2. \quad (\text{B8})$$

Here, we have also used $\langle |\xi_m^n|^2 \rangle = 1$ and that δu_m^n and ξ_m^n are independent, so have zero covariance. At this stage, we can apply Parseval's identity to relate the spatial variance of δu to the Fourier modes

$$\sigma_u^2 = \overline{(\delta u)^2} = \frac{1}{N} \sum_{m=1}^{N-1} (\delta u_m)^2. \quad (\text{B9})$$

Taking the ensemble average of both sides and substituting in Eq. (B8) for each term in the sum in the right-hand side, we can approximate σ_u^2 as its ensemble average, yielding

$$\sigma_u^2 \approx \langle \overline{(\delta u)^2} \rangle = \frac{1}{N} \sum_{m=1}^{N-1} \frac{A}{r_m} \left(1 + \frac{r_m \Delta t}{2} \right)^2. \quad (\text{B10})$$

Solving the system of equations of Eqs. (B3) and (B10) yields estimates for the values of μ_u , μ_v , and σ_u^2 . While the approximations taken to arrive at these results were rather crude, the estimates still retain the key qualitative features and have excellent numerical agreement with the simulated results. Equation (B10) also reveals the dependence of the spread of the activator on the discretization time Δt . For the nonequilibrium-state to agree irrespective of the chosen Δt , a sufficient condition is that $D_u \Delta t/a^2 \ll 1$.

In the continuous time-limit of $D_u \Delta t/a^2 \rightarrow 0$, we find that the variance simplifies to

$$\sigma_u^2 \approx \frac{A}{\sqrt{\kappa(\kappa + 4D_u/a^2)}}, \quad \kappa = \frac{3\mu_u^2 - 2\mu_u\mu_v - 1}{\tau_u}. \quad (\text{B11})$$

Here, κ can be understood as the restoring rate of fluctuations of u away from μ_u . We have also evaluated these quantities in the limit of $N \rightarrow \infty$. Putting things all together, self-consistently solving the system of equations comprised of Eqs. (B3a), (B3b), and (B11) for μ_u , μ_v , and σ_u fully characterizes the nonequilibrium quiescent background.

From Eq. (B11), we see that $\sigma_u^2 \propto A$, meaning that in the absence of noise $\sigma_u = 0$ and μ_u and μ_v revert to their quiescent values of u_0 and v_0 . Gaussian white noise introduces a finite σ_u^2 , which shifts μ_u and μ_v from the quiescent values. For Gaussian white noise to trigger pattern formation, we expect that the integrated noise strength A is sufficiently large such that the spread comparable to the difference between the quiescent values of u_0 and v_0 , namely $\sigma_u \sim (\alpha + \beta - 1)/\alpha$.

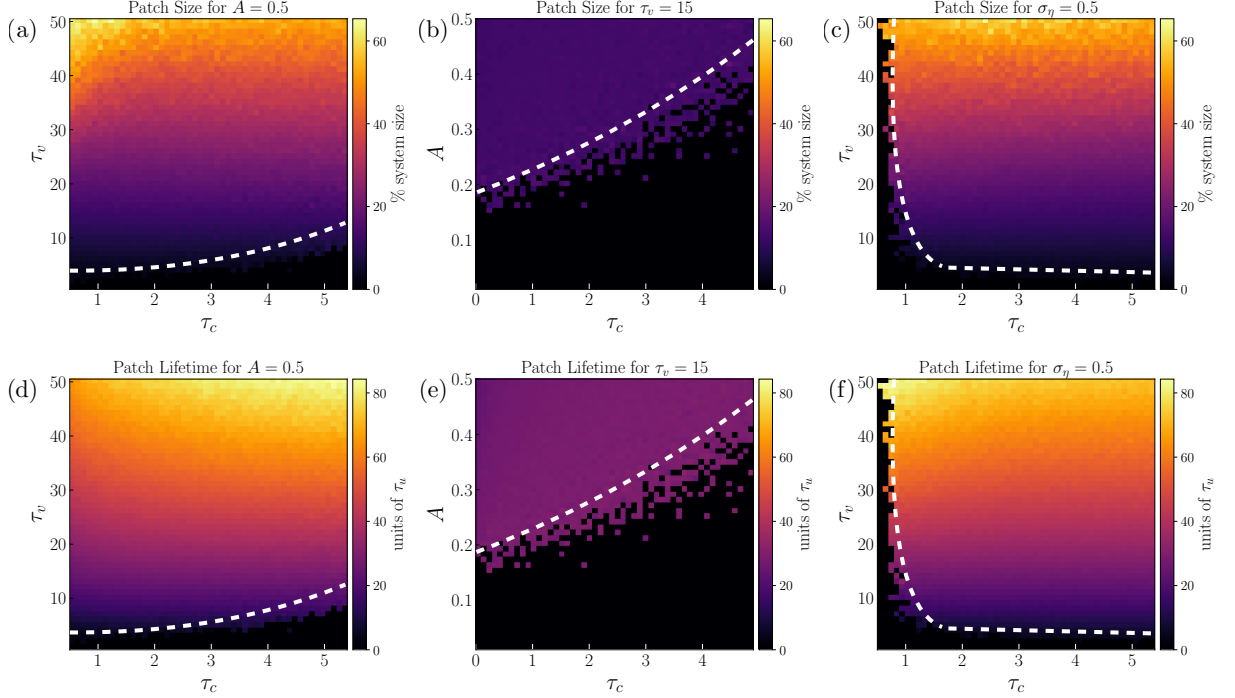


FIG. 6. Properties of the noise-induced excitation patches for the three parameter sweeps corresponding to Fig. 2. Panels (a)–(c) show the characteristic patch size as a percentage of the system size, while panels (d)–(f) show the characteristic patch lifetime in units of τ_u . Panels (a) and (d) vary τ_v and τ_c at fixed integrated noise strength $A = 0.5$; panels (b) and (e) vary A and τ_c at fixed inhibitor response time $\tau_v = 15$; and panels (c) and (f) vary τ_v and τ_c at fixed equal-time rms noise amplitude $\sigma_\eta = 0.5$. The white dashed curves reproduce the dynamical phase boundaries from Fig. 2, and black regions indicate parameter combinations for which no excitation patches were detected. The remaining parameters are $\alpha = 0.5$, $\beta = 0.7$, $D_u = 1$, and $D_v = 5$.

Appendix C: Properties of excited patches

In this Appendix, we examine how the characteristic size and lifetime of the excitation patches vary across the parameter ranges explored in the three phase diagrams of Fig. 2. Figure 6 shows the patch size (top row) and lifetime (bottom row) for the same three parameter sweeps. In the first and third columns, τ_v and τ_c are varied at fixed integrated noise strength A and fixed equal-time rms amplitude σ_η , respectively, while the middle column varies A and τ_c at fixed τ_v .

A common feature of these results is that, once a patch is nucleated, its size and lifetime are governed predominantly by the inhibitor response time τ_v . Both quantities increase approximately linearly with τ_v over much of the pattern-forming regime, consistent with Fig. 5. This agrees with the physical picture that a slower inhibitor allows an excitation to persist longer and spread farther before inhibitory feedback terminates its growth. In particular, when $\tau_c < \tau_v$, the correlation time strongly affects whether pattern formation occurs, as reflected by the phase boundaries, but has comparatively little effect on the size and lifetime of the resulting patches.

The three noise parameterizations further clarify the distinct roles of the stochastic forcing and inhibitor dy-

namics. At fixed A , increasing τ_c shifts the nucleation boundary because increased persistence is accompanied by reduced instantaneous noise amplitude, whereas at fixed σ_η , increasing τ_c strongly promotes nucleation because persistence increases without an amplitude penalty. Despite these opposite effects on the onset of pattern formation, in both cases the size and lifetime of the resulting patches depend predominantly on τ_v . The middle column provides a complementary perspective by fixing τ_v while varying A and τ_c . Here, both noise parameters strongly affect the boundary for patch nucleation, while the characteristic size and lifetime of the patches show a comparatively weaker dependence within the pattern-forming regime.

A different behavior emerges when the noise persists on a timescale comparable to or longer than the inhibitor response time, $\tau_c \gtrsim \tau_v$. This effect is most clearly visible in the patch-lifetime map at fixed σ_η , Fig. 6(f), where the lifetime develops an appreciable dependence on τ_c in the regime of relatively small τ_v and large τ_c . In this regime, a sufficiently strong and long-lived fluctuation can continue to sustain the activator even after the inhibitor has risen to suppress it, thereby prolonging the excitation. Consequently, for $\tau_c \gg \tau_v$, the patch lifetime need no longer be determined primarily by τ_v and can instead become controlled by the persistence time τ_c . Thus, while

the noise primarily controls nucleation for $\tau_c < \tau_v$, suf-

ficiently persistent noise can also directly influence the subsequent dynamics of an already nucleated patch.

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